American Computer Science League

2024-2025 ● Contest 2: Short Problems ● Junior Division

1. Prefix/Infix/Postfix Notation

Convert the following infix expression to prefix:

3 + 4 * (8 − 4 / 2)

A. + 3 4 * − 8 / 4 2
B. + 3 * 4 − 8 / 4 2
C. + 3 * 4 − / 8 4 2
D. + 3 * − 4 8 / 4 2
E. + * 3 4 − 8 / 4 2

2. Prefix/Infix/Postfix Notation

Evaluate the following postfix expression if all numbers are single digits:

9 8 6 2 − * 2 / + 6 3 / −

A. 8
B. 14
C. 20
D. 23
E. 27

3. Bit-String Flicking

Evaluate:

01011 OR NOT 01101

A. 11011
B. 01011
C. 10010
D. 01101
E. 11001

4. Bit-String Flicking

Evaluate:

(LSHIFT-2 01101) OR (RCIRC-3 10110) AND (RSHIFT-1 11011)

A. 01000
B. 10100
C. 11100
D. 01101
E. 11010

5. What Does This Program Do? (Loops)

What is output by the following program?

```
m = 50
n = 40
while n != 0
     r = m % n
     m = n
     n = r
end while
output m
```

A. 50
B. 40
C. 20
D. 10
E. 0

6. What Does This Program Do? (Loops)

What is output by the following program?

```
s = 0
for y = 2 to 8 step 2
     s = s + y / 2
next y
for z = 3 to 1 step -1
     s = s + z ^ 2
next z
output s
```

A. 24
B. 23
C. 22
D. 20
E. 10

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1. 前缀表达式/中缀表达式/后缀表达式

请将下方中缀表达式转换为前缀表达式：

3 + 4 * (8 − 4 / 2)

A. + 3 4 * − 8 / 4 2
B. + 3 * 4 − 8 / 4 2
C. + 3 * 4 − / 8 4 2
D. + 3 * − 4 8 / 4 2
E. + * 3 4 − 8 / 4 2

2. 前缀表达式/中缀表达式/后缀表达式

如果下方所有数字都是一位数，请计算下方后缀表达式：

9 8 6 2 − * 2 / + 6 3 / −

A. 8
B. 14
C. 20
D. 23
E. 27

3. 比特位操作

请计算下方表达式：

01011 OR NOT 01101

A. 11011
B. 01011
C. 10010
D. 01101
E. 11001

4. 比特位操作

请计算下方表达式：

(LSHIFT-2 01101) OR (RCIRC-3 10110) AND (RSHIFT-1 11011)

A. 01000
B. 10100
C. 11100
D. 01101
E. 11010

5. 这个程序是做什么的？（循环）

请问下方程序的输出是什么？

```
m = 50
n = 40
while n != 0
     r = m % n
     m = n
     n = r
end while
output m
```

A. 50
B. 40
C. 20
D. 10
E. 0

6. 这个程序是做什么的？（循环）

请问下方程序的输出是什么？

```
s = 0
for y = 2 to 8 step 2
     s = s + y / 2
next y
for z = 3 to 1 step -1
     s = s + z ^ 2
next z
output s
```

A. 24
B. 23
C. 22
D. 20
E. 10